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LoPresti

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(54) **TREE ROOTING ENCLOSURE APPARATUS**

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(*) Notice: Subject to any disclaimer, the term of this
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U.S.C. 154(b) by 0 days.

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20, 2022.
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A01G 2/20 (2018.01)
A01G 13/10 (2006.01)
(52) **U.S. Cl.**
CPC **A01G 2/20** (2018.02); **A01G 13/10**
(2013.01)
(58) **Field of Classification Search**
CPC H01R 13/51; H01R 13/5208; H01R 13/52
USPC 47/24.1
See application file for complete search history.

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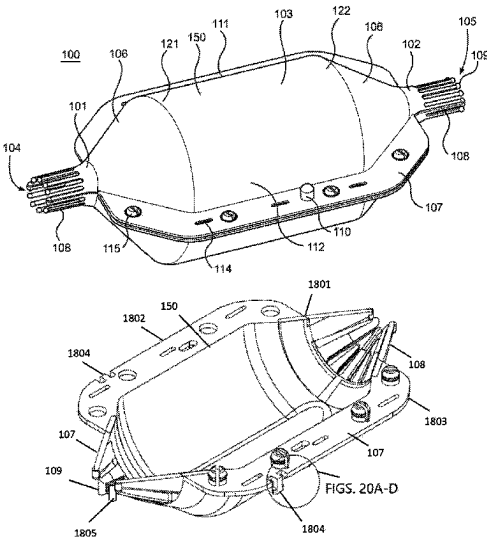
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(74) *Attorney, Agent, or Firm* — Bullock Law; Stephen
Bullock

(57) **ABSTRACT**

A tree rooting enclosure apparatus including a body housing
with a cylindrical body base culminating into a pair of
opposing end openings. The body and end openings are
structured to accommodate a portion of a plant therethrough.
The apparatus as a whole may be made of two removably
engaged elongate halves structured to surround a selected
portion of a plant and attach to itself forming a water-
resistant fit around the selected portion of the plant.

20 Claims, 15 Drawing Sheets



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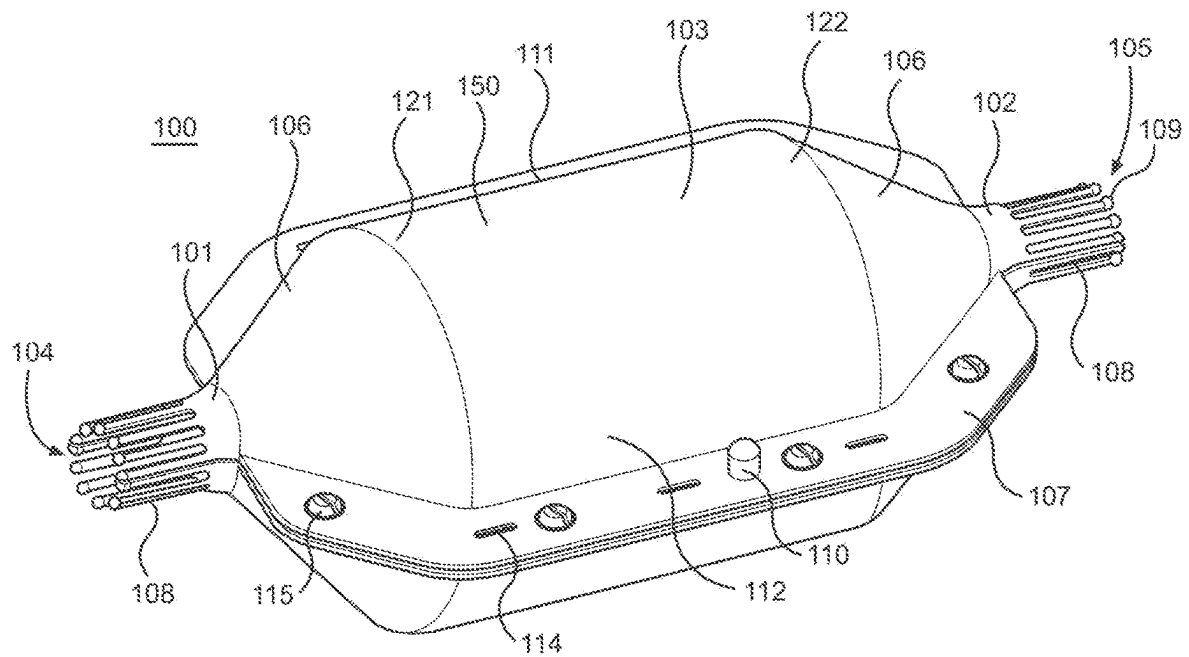


FIG. 1

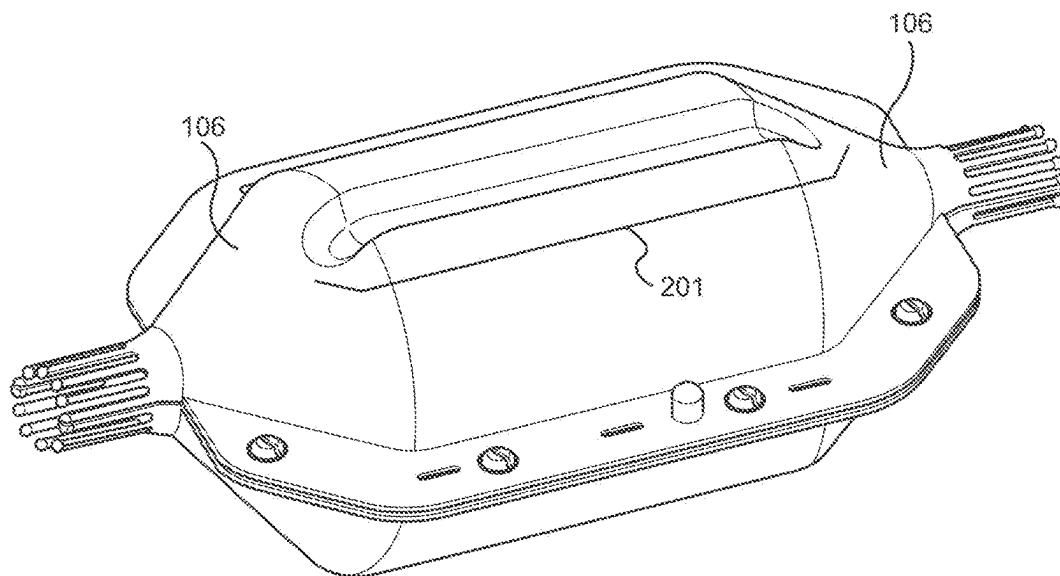


FIG. 2

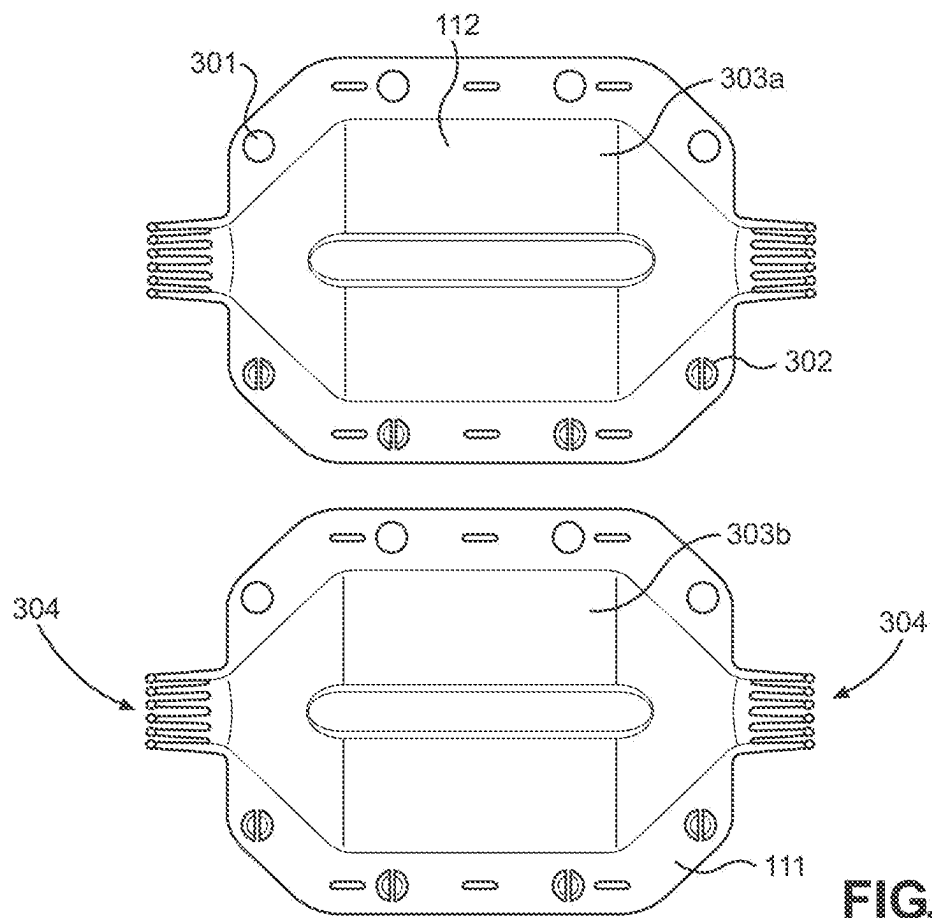


FIG. 3

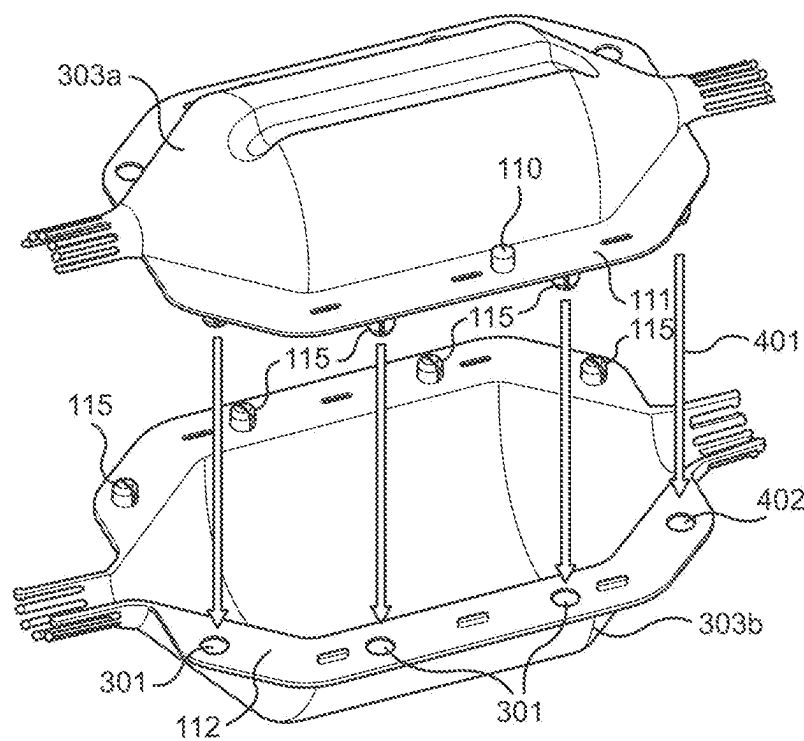


FIG. 4

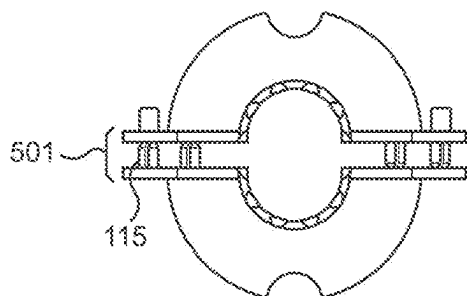


FIG. 5A

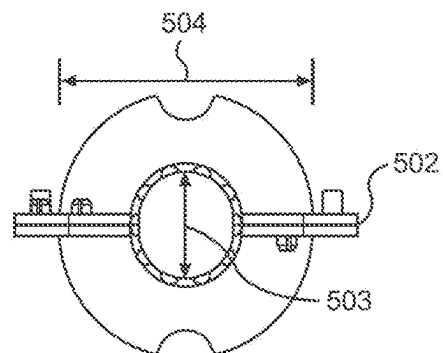


FIG. 5B

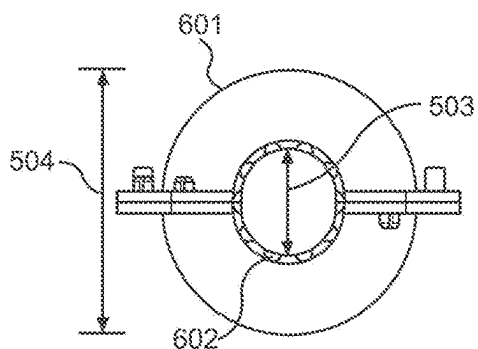


FIG. 6A

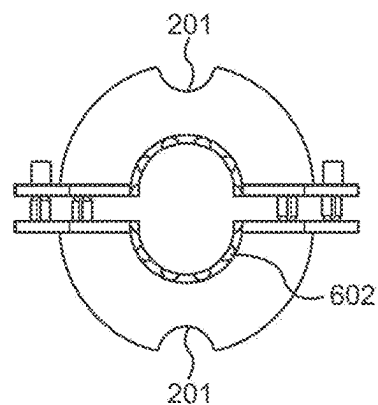


FIG. 6B

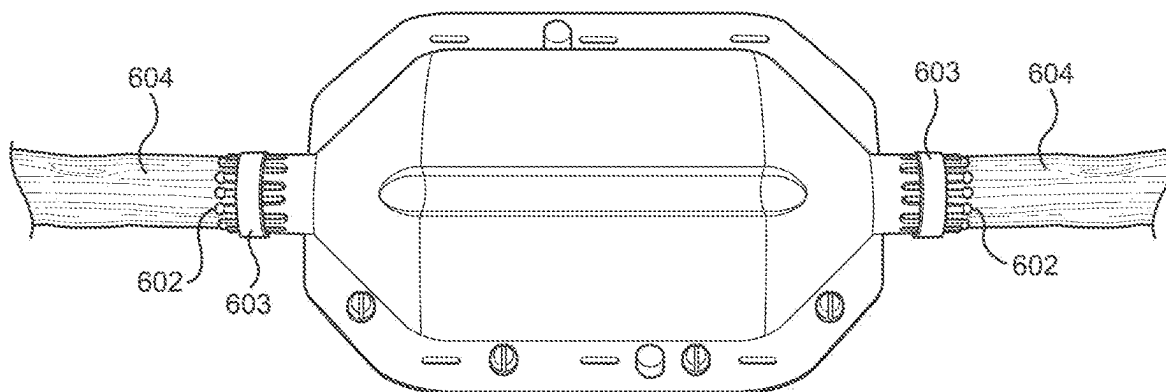


FIG. 6C

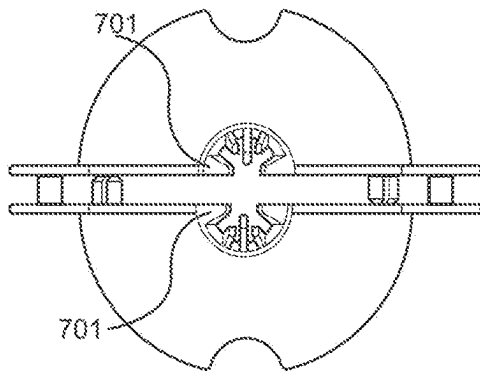


FIG. 7A

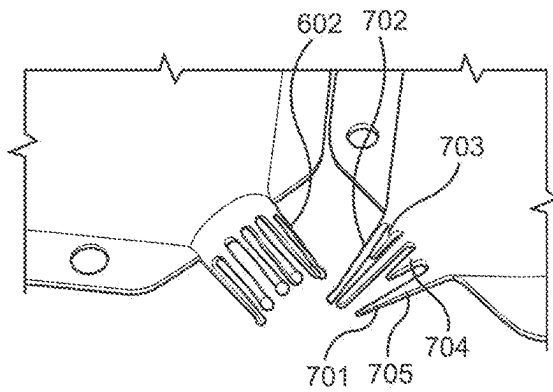


FIG. 7B

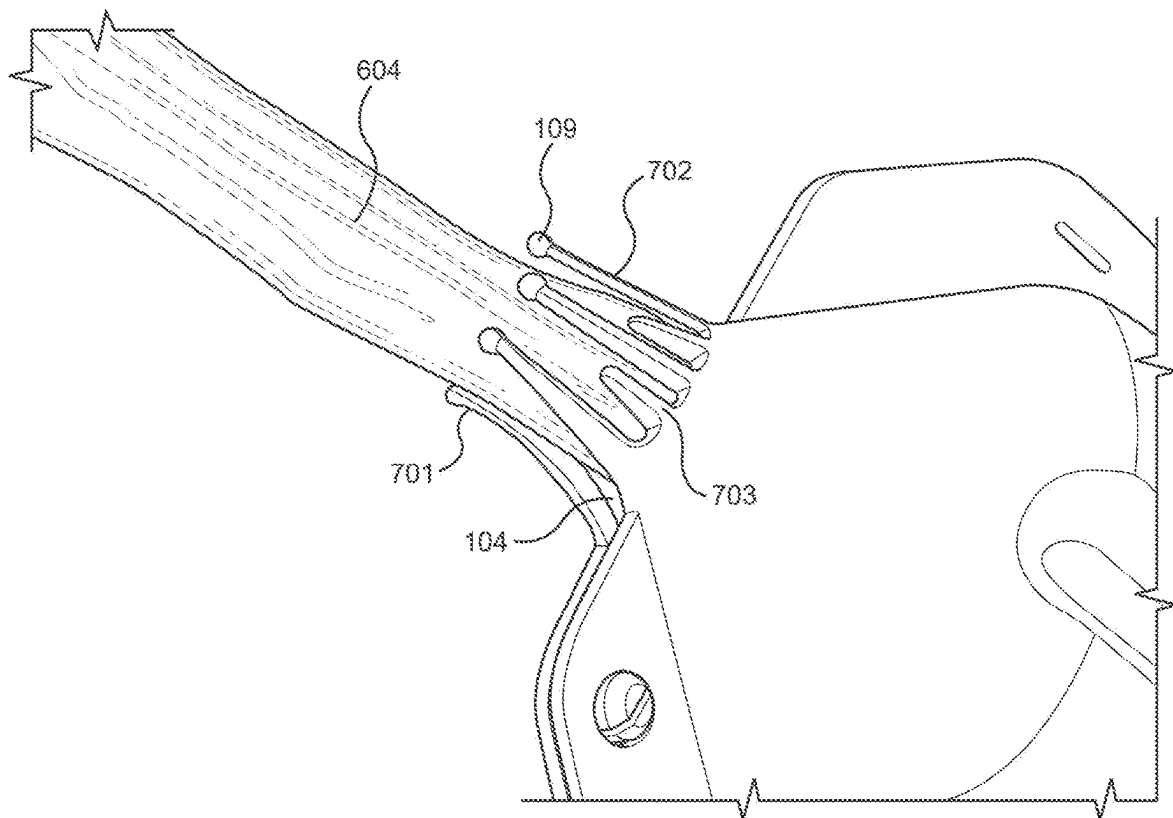


FIG. 7C

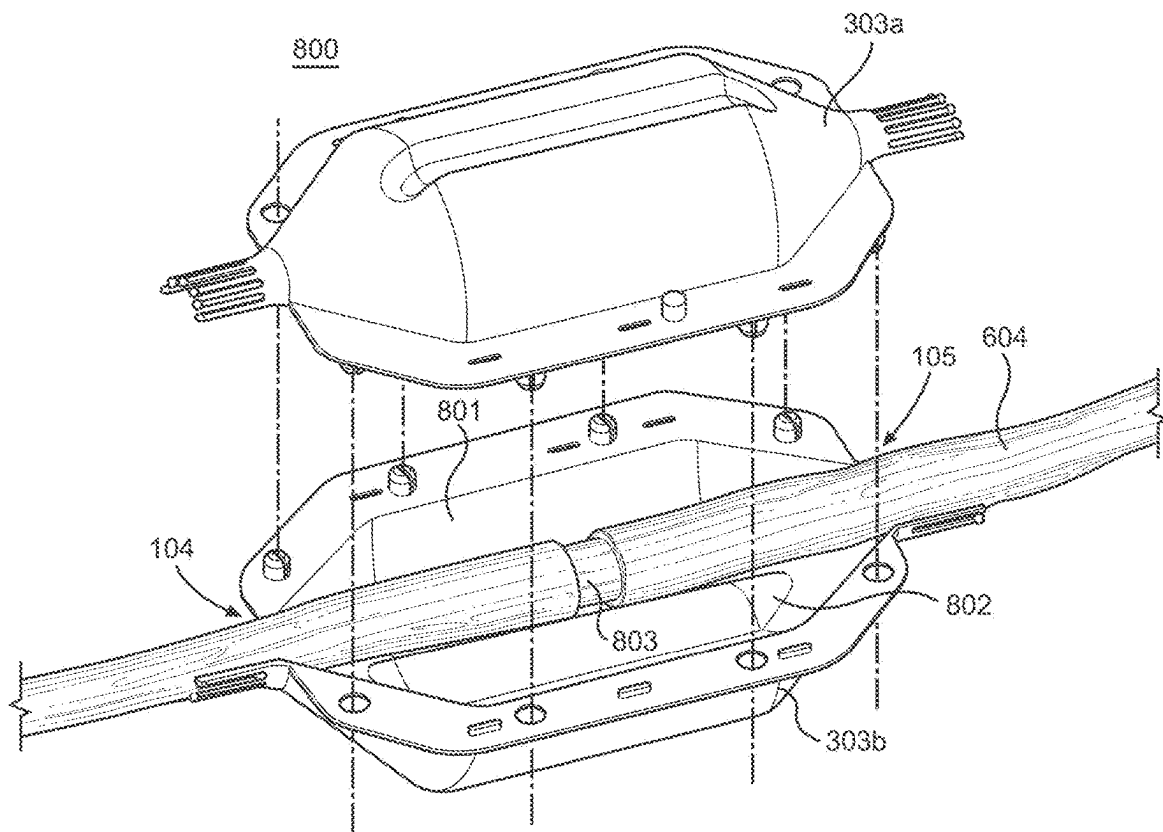


FIG. 8

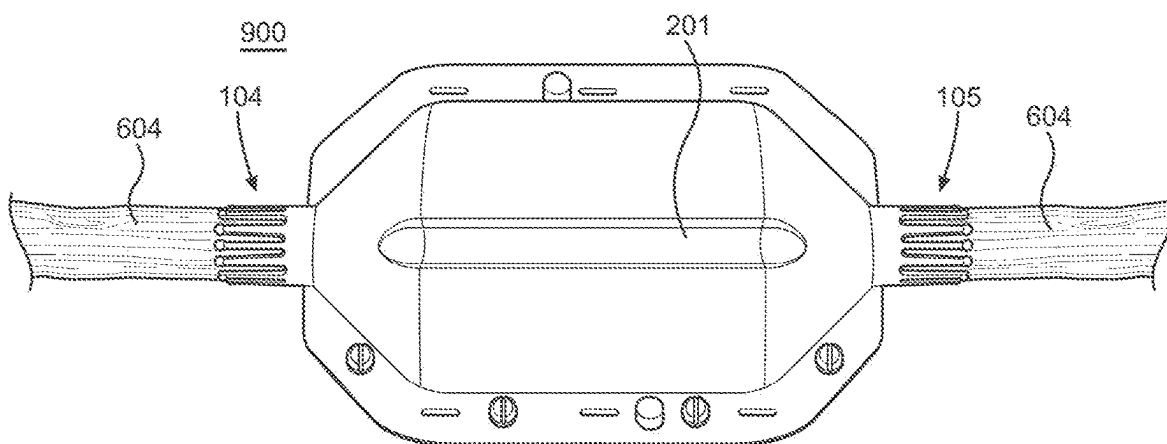


FIG. 9

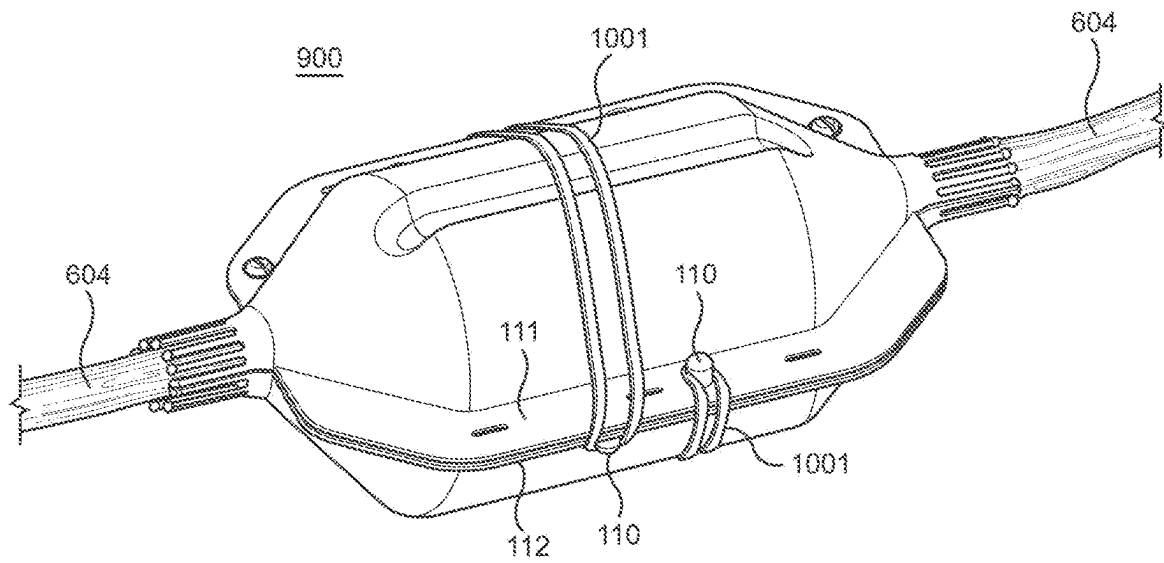


FIG. 10

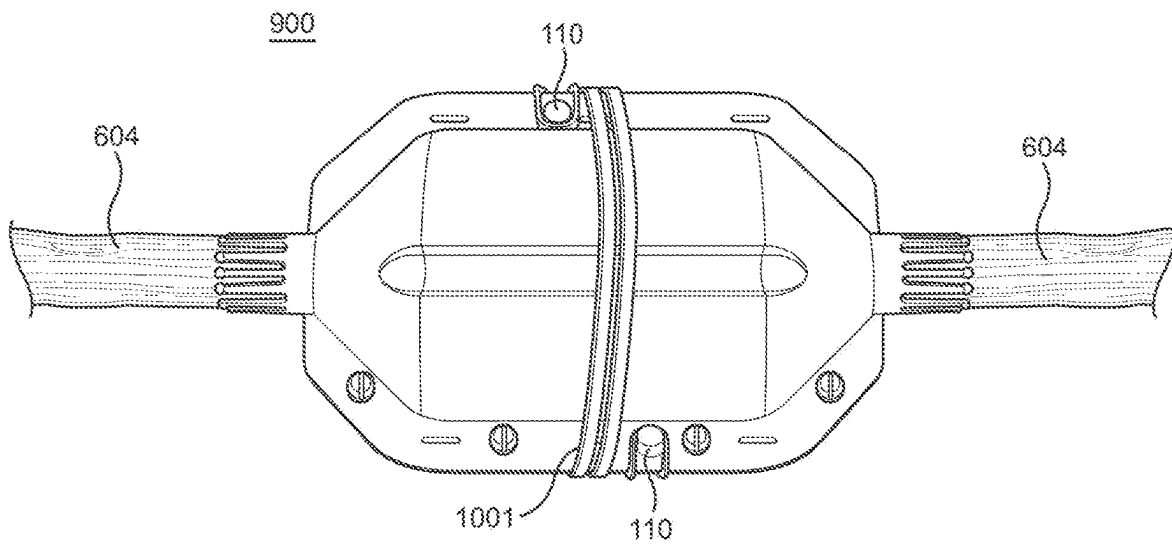


FIG. 11

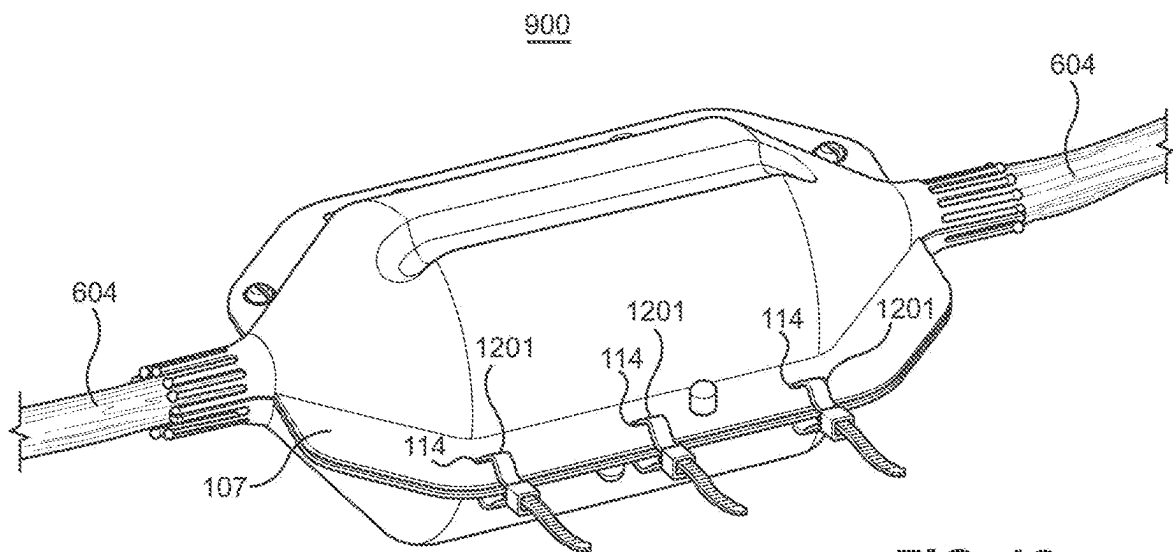


FIG. 12

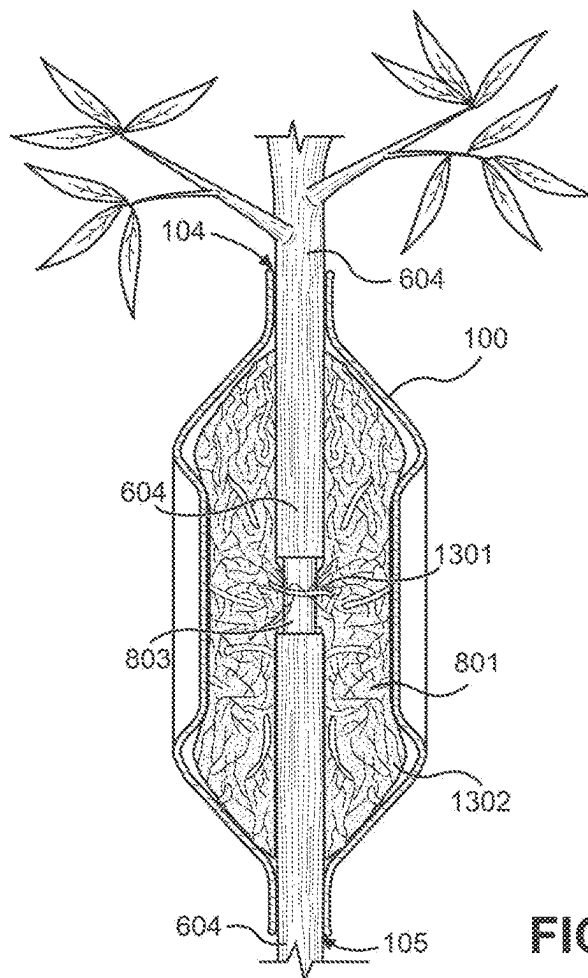


FIG. 13

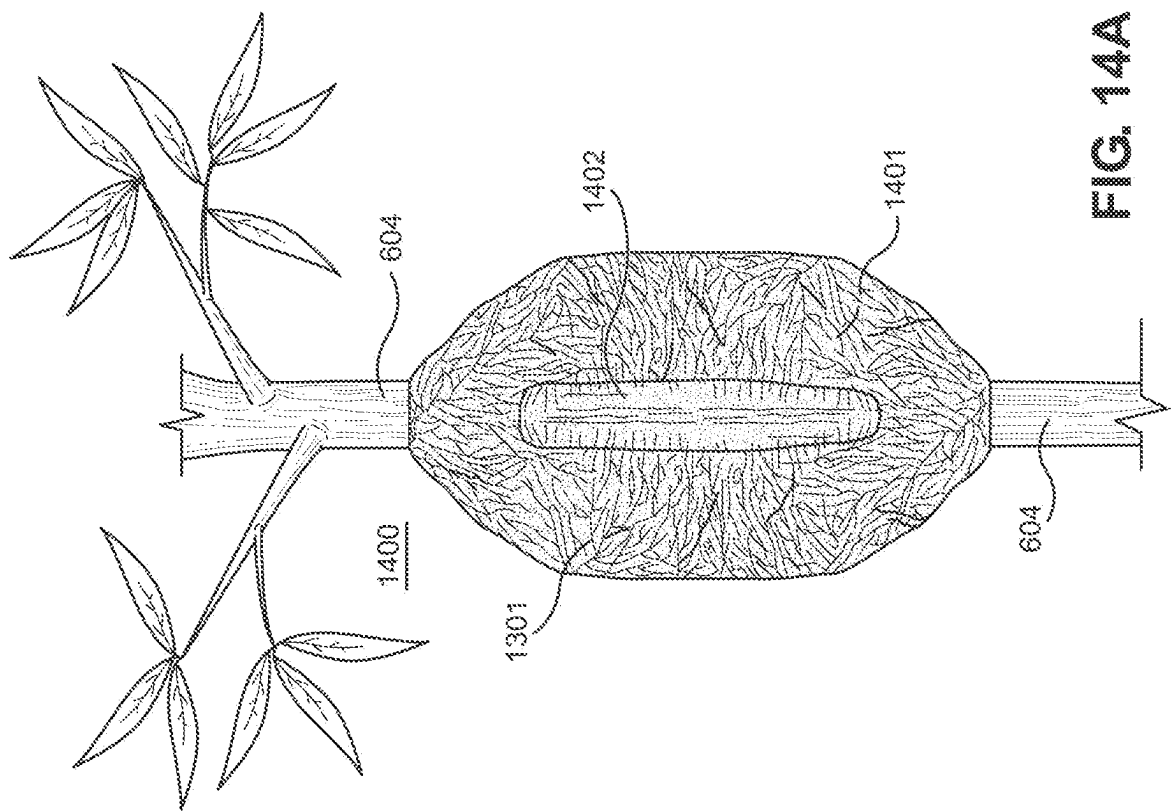


FIG. 14A

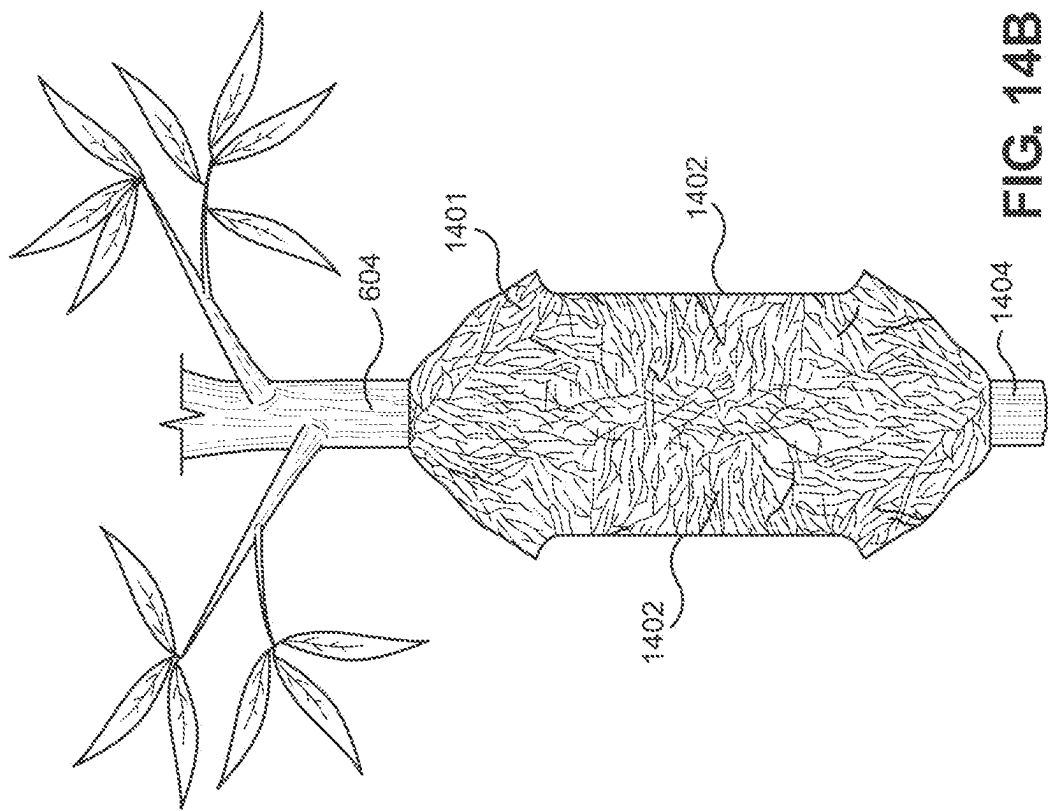


FIG. 14B

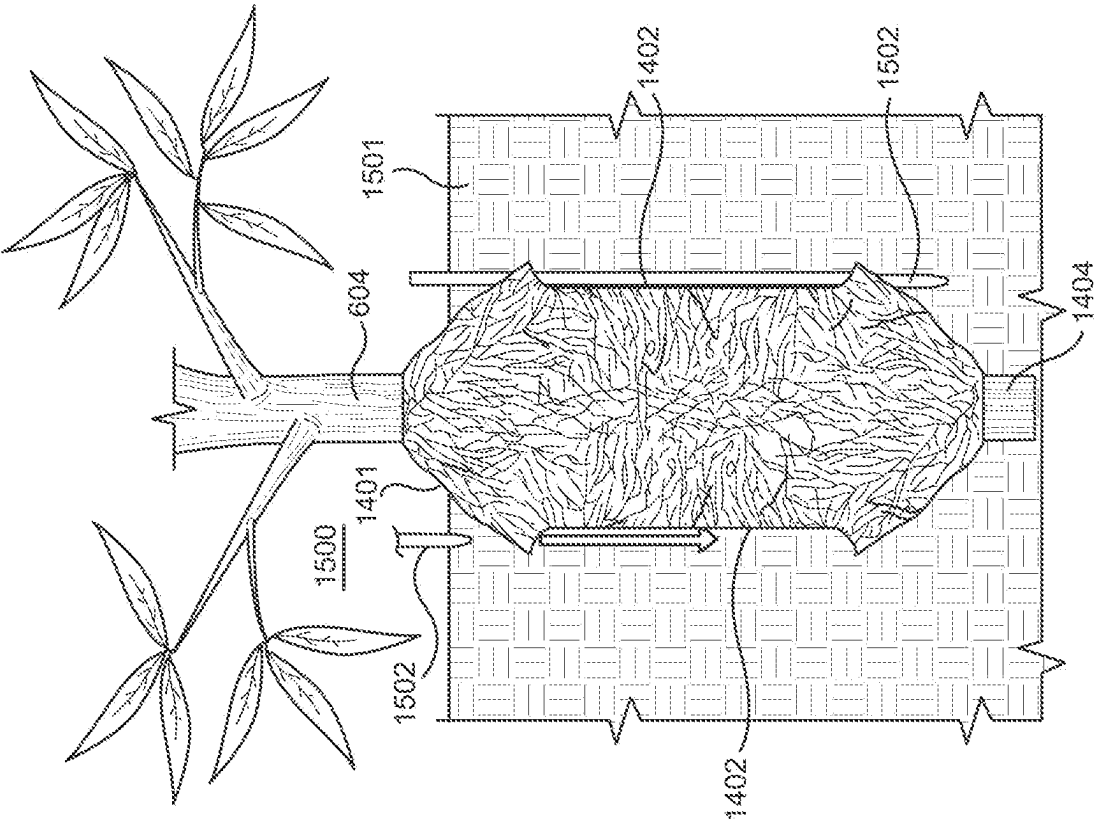


FIG. 15A

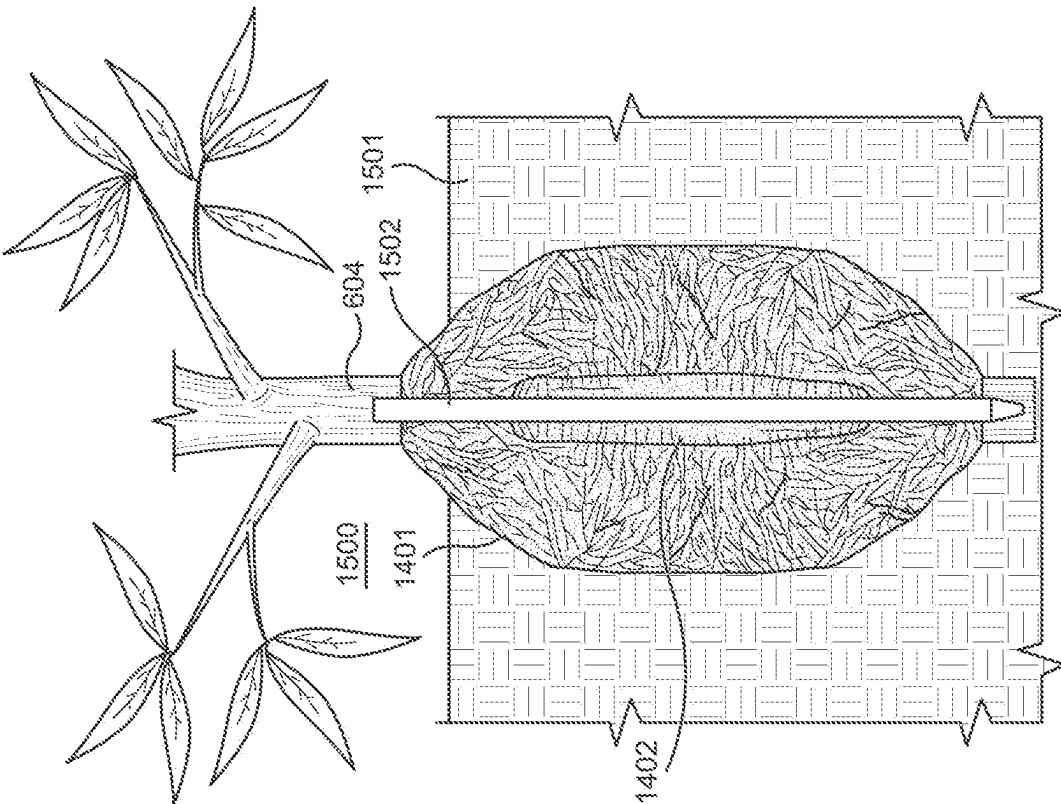


FIG. 15B

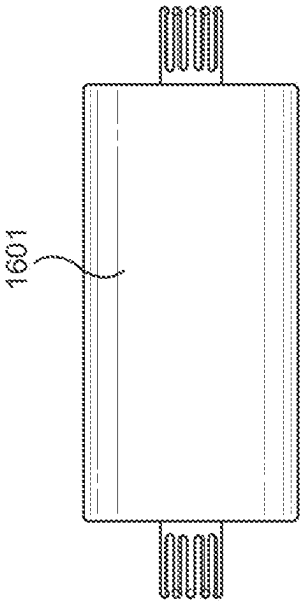


FIG. 16A

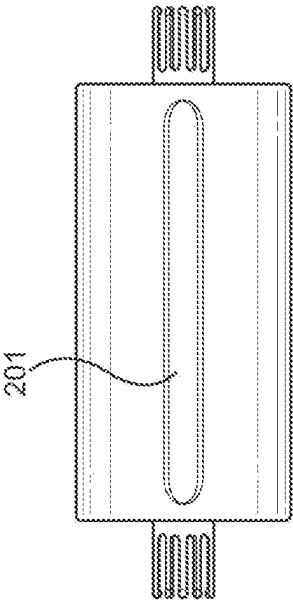


FIG. 16C

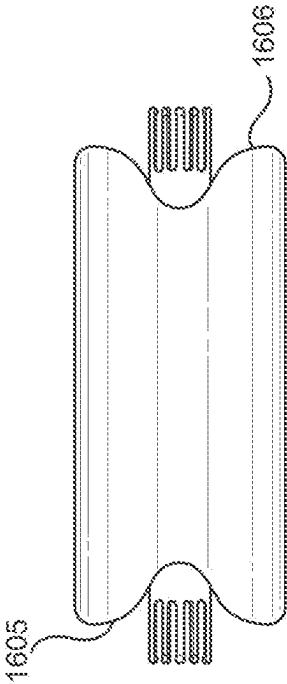


FIG. 16E

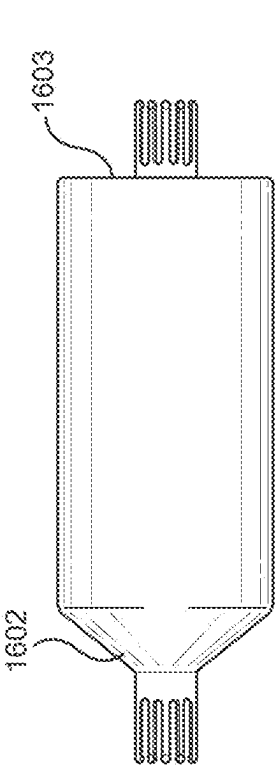


FIG. 16B

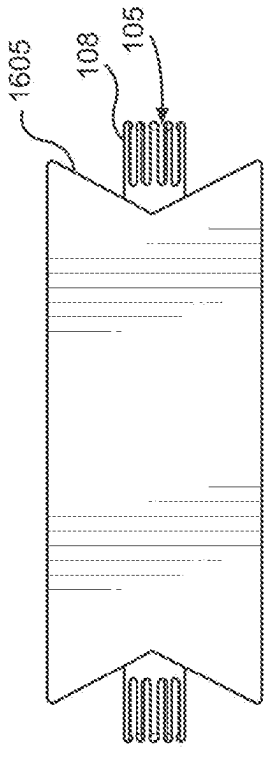


FIG. 16D

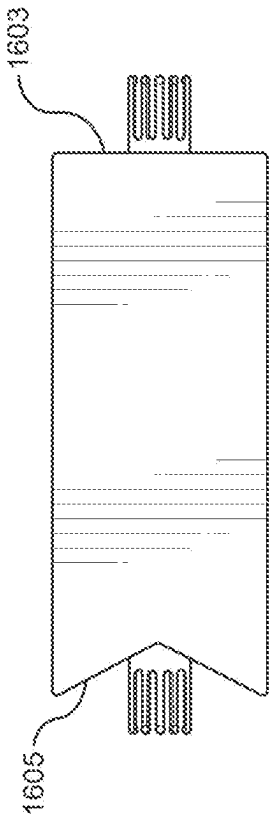


FIG. 16F

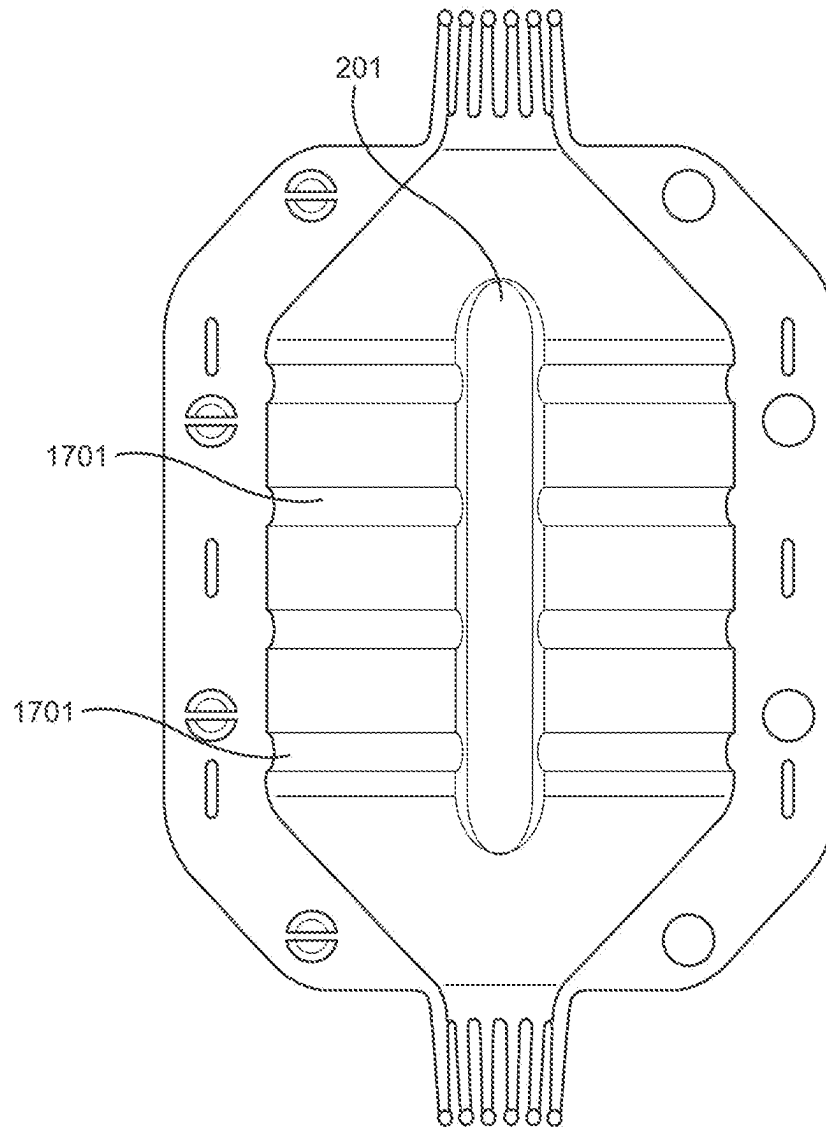


FIG. 17

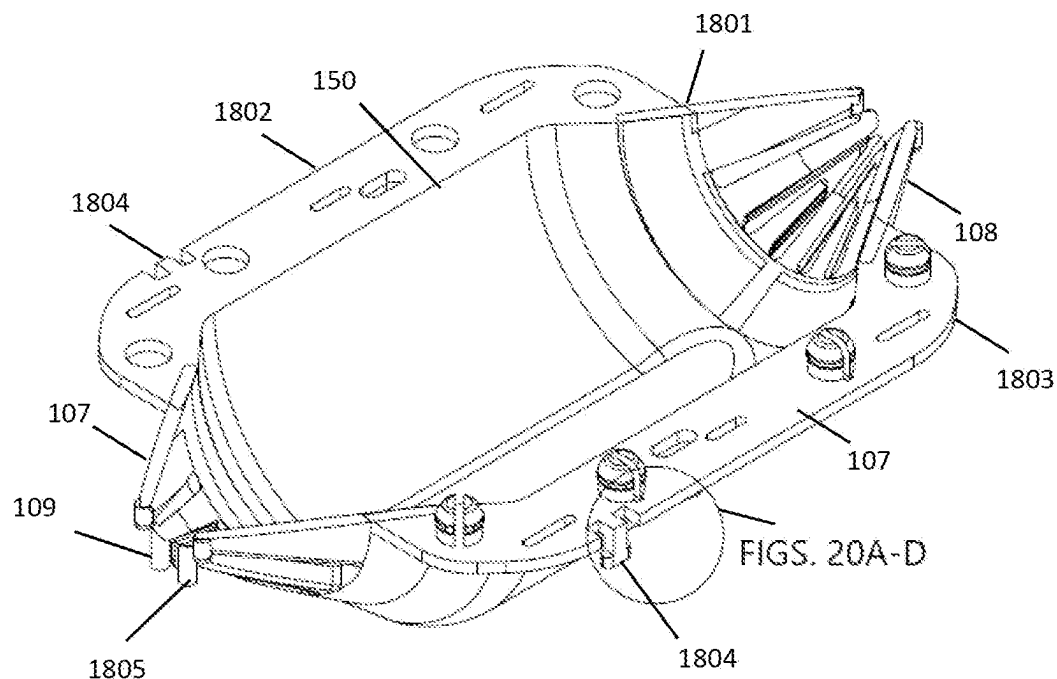


FIG. 18

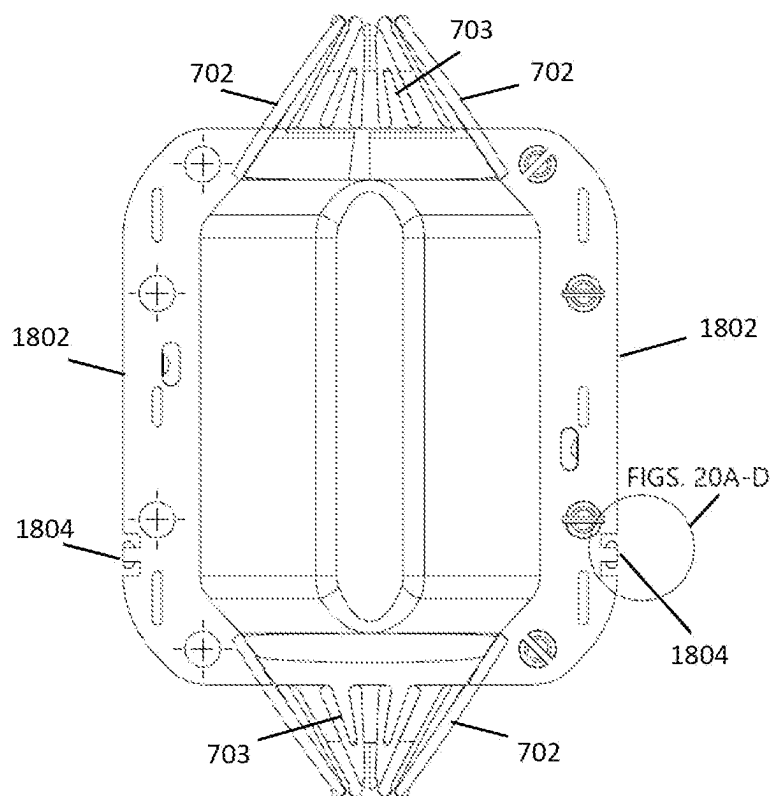


FIG. 19

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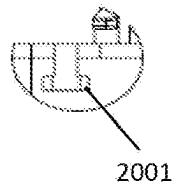


FIG. 20A

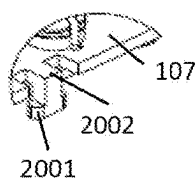


FIG. 20B

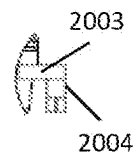


FIG. 20C



FIG. 20D

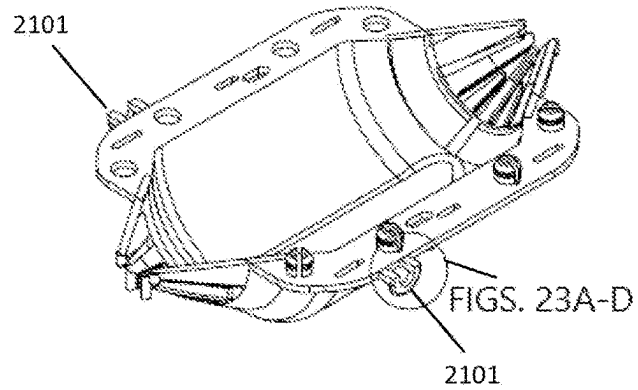


FIG. 21

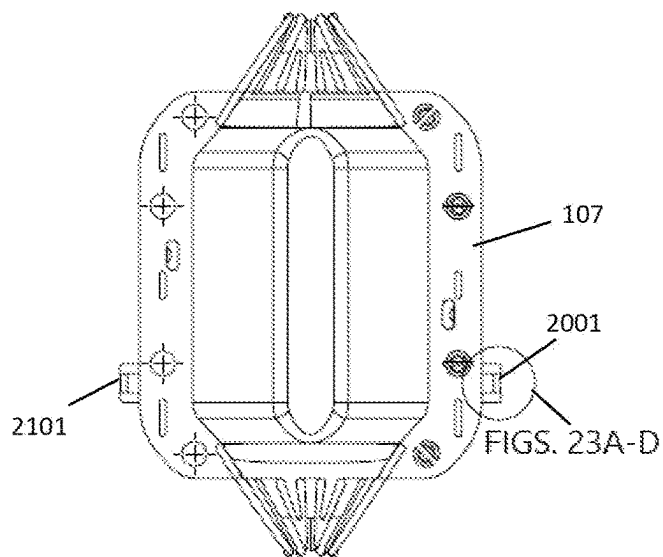


FIG. 22

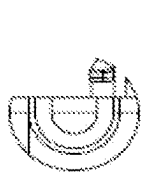


FIG. 23A

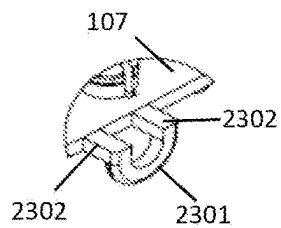


FIG. 23B

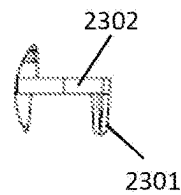


FIG. 23C

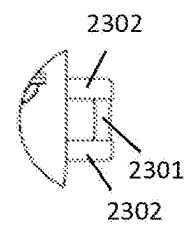


FIG. 23D

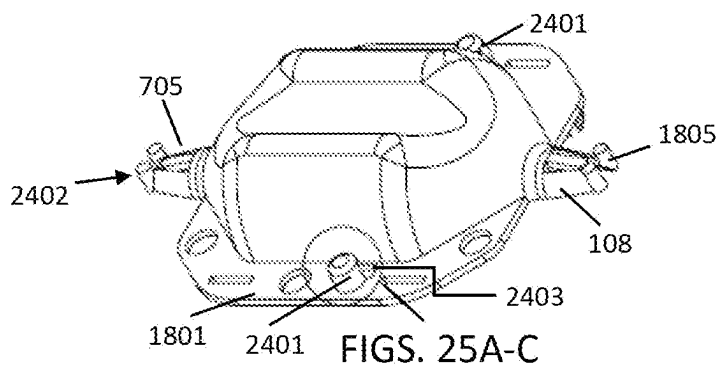


FIG. 24

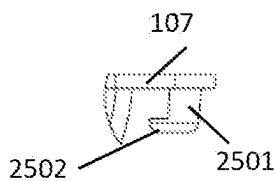


FIG. 25A

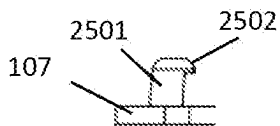


FIG. 25B

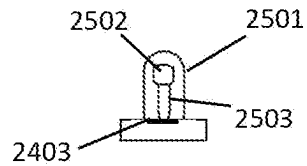


FIG. 25C

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TREE ROOTING ENCLOSURE APPARATUS**FIELD OF THE INVENTION**

The present invention relates to systems and methods for propagating trees. In particular, the present invention relates to growing roots from an artificially created branch aperture enclosed in a container.

BACKGROUND

New trees can be generated from the branches of existing trees by manually creating a fissure or wound on a tree branch, which removes or injures a plant's phloem layer. The injury interrupts the flow of sugars and photosynthates to the roots, which then accumulate at the injury site. This accumulation when packed within a conducive medium generates root producing hormones. Therefore, enclosing the fissure with a wrapping filled with sphagnum moss or a potting mix facilitates the stimulation of adventitious root buds rather than leaves. Once a sufficient amount of time has passed, a mature network of roots may form to allow the branch to stand alone as a separate tree. The tree branch may then be removed from its original tree at a point below the newly formed network of roots and the removed branch may be planted and nurtured as a new tree.

There exists a need for a rooting enclosure that efficiently and effectively envelops a manually created tree fissure when propagating trees. The enclosure should be optimally shaped to mimic the natural shape of the branch and should provide a number of fastening and stabilizing options for securing the enclosure that is resistant to water and animals. The enclosure should be reusable and modular in nature. Therefore, there exists a need in the art for a tree rooting enclosure apparatus and associated methods.

SUMMARY OF THE INVENTION

Embodiments of the present invention are related to a tree rooting enclosure apparatus including a body with a cylindrical body base, a pair of opposing end openings, and a plurality of spindles extending outward from the exterior edges of the pair of opposing end openings. The apparatus may be made of two removably engageable halves, each half being an identical duplicate of the other. Each half may be structured to mate with the other half when each respective half is inversely oriented with respect to the other half. Both halves when engaged may envelop at least a portion of a tree branch to propagate a rooting system.

In this embodiment, the plurality of spindles may include at least one spindle shorter in length than the remaining spindles and the spindles may be structured in at least one of a symmetrically splayed orientation and an angled orientation. The cylindrical body base diameter may be proportional to the end opening diameters by a ratio of 1 to 0.75 respectively and the cylindrical body base may include a perimeter tab extending distally therefrom. The perimeter tab may include an elbow shaped T-lever structured to facilitate opening the apparatus when both halves are engaged. Furthermore, the perimeter tab may include four rounded corners and a pair of parallel sides and the pair of opposing end openings may form a concave void in the perimeter tab at each respective end of the apparatus.

Another embodiment may include a tree rooting enclosure apparatus with a body including a cylindrical body base, a perimeter tab extending from the cylindrical body base, a pair of opposing end openings, a plurality of spindles

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extending outward from the exterior edges of the pair of opposing end openings, and a plurality of longitudinal channels extending the length of the cylindrical body base. The apparatus may be made of two removably engageable halves, each half being an identical duplicate of the other. Each half may be structured to mate with the other half when each respective half is inversely oriented with respect to the other half and both halves when engaged may envelop at least a portion of a tree branch to propagate a rooting system.

In this embodiment, the plurality of spindles on each half may include at least one pair of longer spindles flanking at least one shorter medial spindle. Additionally, the opposing sides of the perimeter tab may include a pair of fastening hooks structured to facilitate securing the tree rooting enclosure apparatus in a closed position with a perimeter fastening strap. Furthermore, at least one of the plurality of spindles may include an end distal to the end opening that is terminated by a cylinder. The perimeter tab of each half may include a concave protrusion structured to facilitate a user pulling apart the two halves when the two halves are engaged. The concave protrusion may be dimensioned to allow a user's finger to fit into the concave void and the perimeter tab may include a plurality of fastening apertures structured to accommodate a plurality of side cinches for securing the apparatus in a closed position.

The perimeter tab may include at least one side with a plurality of joinder pegs and an opposing side with a plurality of joinder holes. The plurality of joinder pegs of the first half enclosing member may be structured to removably engage the plurality of joinder holes of the second half enclosing member to form a water-resistant fitting when the apparatus is in a closed position. The plurality of spindles may be in an angled orientation and may include at least one spindle shorter in length than the others. At least one spindle may include an end distal to the end opening that is terminated by a cylinder.

Another embodiment may include a tree rooting enclosure apparatus with a body including a cylindrical body base body with a perimeter tab extending the entire longitudinal length of two sides of the cylindrical body base with at least one of a concave finger tab extension and an elbow shaped T-lever. It may include a pair of opposing end openings and a plurality of spindles extending outward from the exterior edges of the pair of opposing end openings. The apparatus may be made of two removably engageable halves, each half being an identical duplicate of the other and each half may be structured to mate with the other half when each respective half is inversely oriented with respect to the other half. Both halves when engaged may envelop at least a portion of a tree branch to propagate a rooting system. Furthermore, the concave finger tab protrusion and elbow shaped T-lever may be structured to facilitate a user pulling apart the two halves when the two halves are engaged.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a top perspective view of a tree rooting enclosure apparatus according to an embodiment of the invention.

FIG. 2 is a top perspective view of the tree rooting enclosure apparatus according to another embodiment of the invention.

FIG. 3 is a top view of components of the tree rooting enclosure apparatus according to an embodiment of the invention.

FIG. 4 is a side perspective view of components of the tree rooting enclosure apparatus being fitted together according to an embodiment of the invention.

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FIG. 5A is a lateral end view of the tree rooting enclosure apparatus according to an embodiment of the invention.

FIG. 5B is a lateral end view of the tree rooting enclosure apparatus according to an embodiment of the invention.

FIG. 6A is a lateral end view of the tree rooting enclosure apparatus according to an embodiment of the invention.

FIG. 6B is a lateral end view of the tree rooting enclosure apparatus according to an embodiment of the invention.

FIG. 6C is a top perspective view of the tree rooting enclosure apparatus with accessory components according to an embodiment of the invention.

FIG. 7A is a lateral end view of the tree rooting enclosure apparatus according to another embodiment of the invention.

FIG. 7B is a segmented view comparing different end embodiments of the tree rooting enclosure apparatus.

FIG. 7C is a segmented environmental perspective view of one end of the tree rooting enclosure apparatus according to an embodiment of the invention.

FIG. 8 is a perspective environmental view of a tree rooting enclosure apparatus in an opened position according to an embodiment of the invention.

FIG. 9 is a top perspective environmental view of a tree rooting enclosure apparatus in a closed position according to an embodiment of the invention.

FIG. 10 is a side perspective environmental view of the tree rooting enclosure apparatus with accessory components according to an embodiment of the invention.

FIG. 11 is a top perspective environmental view of the tree rooting enclosure apparatus with accessory components according to an embodiment of the invention.

FIG. 12 is a side perspective environmental view of the tree rooting enclosure apparatus with accessory components according to an embodiment of the invention.

FIG. 13 is a cross sectional environmental view of the tree rooting enclosure apparatus according to an embodiment of the invention.

FIG. 14A is an environmental view of a root network mass created by the tree rooting enclosure apparatus according to an embodiment of the invention.

FIG. 14B is an environmental view of a root network mass created by the tree rooting enclosure apparatus according to an embodiment of the invention.

FIG. 15A is a side environmental view of a root network mass created by the tree rooting enclosure apparatus with a stabilizing stake fitted thereto.

FIG. 15B is a front environmental view of a root network mass created by the tree rooting enclosure apparatus with a pair of stabilizing stakes fitted thereto.

FIG. 16A illustrates a different embodiment of the tree rooting enclosure apparatus.

FIG. 16B illustrates a different embodiment of the tree rooting enclosure apparatus.

FIG. 16C illustrates a different embodiment of the tree rooting enclosure apparatus.

FIG. 16D illustrates a different embodiment of the tree rooting enclosure apparatus.

FIG. 16E illustrates a different embodiment of the tree rooting enclosure apparatus.

FIG. 16F illustrates a different embodiment of the tree rooting enclosure apparatus.

FIG. 17 illustrates another embodiment of the tree rooting enclosure apparatus.

FIG. 18 is a perspective view of one half of the tree rooting enclosure apparatus according to an embodiment of the invention.

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FIG. 19 is a top view of one half of the tree rooting enclosure apparatus according to the embodiment of the invention illustrated in FIG. 18.

FIG. 20A is a front view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 18.

FIG. 20B is a perspective view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 18.

FIG. 20C is a side view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 18.

FIG. 20D is a top view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 18.

FIG. 21 is a perspective view of one half of the tree rooting enclosure apparatus according to an embodiment of the invention.

FIG. 22 is a top view of one half of the tree rooting enclosure apparatus according to the embodiment of the invention illustrated in FIG. 21.

FIG. 23A is a front view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 21.

FIG. 23B is a perspective view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 21.

FIG. 23C is a side view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 21.

FIG. 23D is a top view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 21.

FIG. 24 is a perspective view of one half of the tree rooting enclosure apparatus according to an embodiment of the invention.

FIG. 25A is a side view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 24.

FIG. 25B is a side view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 24.

FIG. 25C is a front view of componentry of the tree rooting enclosure apparatus illustrated in FIG. 24.

DETAILED DESCRIPTION OF THE INVENTION

The present invention will now be described in detail with reference to the accompanying drawings. The embodiment descriptions are illustrative and not intended to be limiting in any way. Other embodiments of the invention will readily suggest themselves to persons with ordinary skill in the art after having the benefit of this disclosure. Accordingly, the following embodiments are set forth without any loss of generality and without imposing limitation upon the claimed invention.

Directional terms such as “above” “below” “upper” “lower” and other like terms are used for the convenience of the reader in reference to the drawings. Additionally, the description may contain terminology to convey position, orientation, and direction without departing from the principles of the present invention. Such positional language should be taken in context of the represented drawings.

Quantitative terms such as “generally” “substantially” “mostly” and other like terms are used to mean that the referred object, characteristic, or quality constitutes a majority of the referenced subject. Likewise, use of the terms such as first and second do not necessarily designate a limitation of quantity. Such terms may be used as a method of describing the presence of at least one of the referenced elements or may provide a means of differentiating orientation. The meaning of any term within this description is dependent upon the context within which it is used, and the meaning may be expressly modified.

Referring now to FIG. 1, the tree rooting enclosure apparatus 100, hereinafter the apparatus 100, is shown. The

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apparatus 100 may include an apparatus first end 101, an apparatus second end 102, a body 150 with cylindrical body base 103 and a pair of body tapers 106 at a body first end 121 and a body second end 122. The apparatus 100 may also include what may be designated as an apparatus right side 111 and an apparatus left side 112.

The body tapers 106 may be a pair of opposing tapered ends culminating into a pair of opposing end openings 104, 105. As such, there may be a first end opening 104 and a second end opening 105. In some embodiments, the opposing end openings 104, 105 may include a diameter relative to the diameter of the cylindrical body base 103. By way of non-limiting example, this may be a cylindrical body base diameter to end opening diameter ratio of 2.5 to 1 respectively. Furthermore, extending distally from the edges of the end openings 104, 105 may be a plurality of spindles 108. In this particular embodiment, the spindles 108 are shown having equivalent length. Additionally, in some embodiments, the spindles 108 may include spindle ends 109 that may be bulbous.

The apparatus 100 may also include a perimeter tab 107 that may extend distally and surround the cylindrical body base 103 and body tapers 106 along a longitudinal axis of the apparatus 100. In some embodiments the perimeter tab 107 may be polygonal and structured as an octagonal lip with rounded vertices. In fact, as will be described more fully hereafter, the apparatus 100 may be constructed from two congruent enclosing members each comprising a longitudinal half of the apparatus 100 when bifurcated along its longitudinal medial axis. Therefore, as will be described, the apparatus 100 as a whole is structured from two removably engaged halves forming a water-resistant fit. It should be noted that the water-resistant fit also protects against intrusion from animals such as birds, squirrels, and the like.

The perimeter tab 107 may extend around the perimeter of each enclosing member. As such, the perimeter tab 107 may include a plurality of joinder pegs 115 extending distally from the interior right underside of the perimeter tab 107. Furthermore, the exterior of the perimeter tab 107 may also include a plurality of fastening dowels 110 protruding therefrom. In this embodiment, there are two fastening dowels 110 asymmetrically positioned in a staggered formation on the apparatus right side 111 and apparatus left side 112. Additionally, the perimeter tab 107 may include a plurality of fastening apertures 114, which in some embodiments may be slits or holes for tying the apparatus 100 together at its midsection.

FIG. 2 illustrates that in some embodiments the apparatus 100 may include a plurality of longitudinal channels 201 that may extend the length of the cylindrical body base 103. A longitudinal channel 201 may be a concave indentation in the exterior of the cylindrical body base 103. In this particular embodiment, there may be two longitudinal channels 201 on opposing surfaces of the apparatus 100. One skilled in the art will appreciate that there may be more or less longitudinal channels 201 within the apparatus 100 depending on need, preference, and circumstance. Likewise, in some embodiments, the longitudinal channels 201 may extend past the cylindrical body base 103 into the body taper 106 and in some embodiments may create an indented arch therein.

FIG. 3 illustrates that the apparatus 100 may be made of two congruent enclosing members 303a, 303b structured to removably engage one another. Further shown are a plurality of joinder holes 301 on the apparatus left side 112 sized and positioned to accommodate the plurality of joinder pegs 115 on the apparatus right underside 111. As such, the perimeter tab 107 may include at least one side with a plurality of

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joinder pegs 115 and an opposing side with a plurality of joinder holes 301. In some embodiments, the plurality of joinder pegs 115 may include an exterior 302 with a facilitating groove to assist with positioning and fitting the joinder pegs 115 into and out of the joinder holes 301 with a friction fit. Furthermore, the opposing end openings 104, 105 may be formed from semi-circular medial openings 304 at opposing ends of each enclosing member 303a, 303b.

FIG. 4 shows how the two half enclosing members 303a, 303b may be removably engaged. As shown, each enclosing member 303a, 303b is an identical duplicate of the other. A first enclosing member 303a is positioned in an inverse orientation to a second enclosing member 303b. As such, the plurality of joinder pegs 115 on the right side 111 of the first enclosing member 303a are aligned with the plurality of joinder holes 301 on the left side 112 of the second enclosing member 303b. A user may removably engage the first and second enclosing members 303a, 303b by exerting force 401 to combine the joinder pegs 115 and joinder holes 301 with a friction fit thereby placing the apparatus 100 in a closed position.

FIGS. 5A and 5B are lateral end views of the apparatus 100. As shown, the joinder pegs 115 are aligned with the joinder holes 301 and the enclosing member halves 303a, 303b are stacked vertically. FIG. 5B shows that when fitted together, the two enclosing member halves 303a, 303b create a water-resistant fit 502 at the perimeter tab 107 when the apparatus 100 is in a closed position. Also shown is the comparison between the cylindrical body base diameter 504 and an end opening diameter 503. As such, by way of non-limiting example, the ratio between the cylindrical body base diameter 504 at its widest point and an end opening diameter 503 may be 2.5 to 1 respectively in the closed position. However, one skilled in the art will appreciate that the ratio may be slightly larger or slightly smaller depending on need, preference, and circumstance.

FIGS. 6A and 6B are lateral end views of the apparatus 100 comparing two slightly different embodiments, but with similar spindle 108 configurations. In one embodiment, as shown in FIG. 6A, the apparatus 100 may include a smooth body base 601. Meaning, this embodiment does not include the longitudinal channel 201 undulating the body base 103. However, the ratio between the cylindrical body base diameter 504 at its widest point and an end opening diameter 503 may still be 2.5 to 1 respectively. But, similar to the previous embodiment, one skilled in the art will appreciate that the ratio may be larger or smaller depending on need, preference, and circumstance. In fact, the entire apparatus 100 may be made larger or smaller to accommodate differently sized plants and branches. By way of non-limiting example, the apparatus 100 as a whole may be much smaller making the end openings 104, 105 much smaller. Likewise, the apparatus 100 may be made much larger to accommodate larger branches and plants whereby the end openings 104, 105 and cylindrical body base diameter 504 would be much larger. Likewise, one skilled in the art will appreciate that the ratio between the end opening diameter 503 and the cylindrical body base diameter 504 may be larger or smaller depending on need, preference, and circumstance.

FIG. 6B illustrates by way of comparison against FIG. 6A that in some embodiments the apparatus 100 may include a longitudinal channel 201 thereby undulating the body base 103. However, both FIGS. 6A and 6B show an embodiment whereby the spindles 108 are in a symmetrically splayed configuration 602. This means that the spindles 108 are level

with, and evenly distributed around the edge of the end opening 104. In this embodiment the spindles 108 are also of the same length.

FIG. 6C illustrates how the apparatus 100 may be secured when in a closed position around a tree branch 604 with symmetrically splayed spindles 602. As shown, the spindles 108 may fit around the tree branch 604 at each end opening 104, 105 and may be tightened therearound by at least one end cinch 603 at each. By tightening the end cinch 603 around the spindles 108, the apparatus 100 in turn is more firmly secured to a tree branch 604 at a desired location. However, one skilled in the art will appreciate that the end cinch 603 may or may not be used to supplement securement of the apparatus 100 depending on need, preference, and circumstance.

FIG. 7A is a lateral end view of the apparatus 100 showing an embodiment with angled spindles 701. In this embodiment, the angled spindles 701 are biased toward the center of the end openings 104, 105 creating a more fitted mount on a tree branch 604. Here, the angled spindles 701 narrow the end openings 104, 105 so that when a tree branch 606 is fitted therethrough, the angled spindles 701 are already secured thereto without the need for end cinches 603.

FIG. 7B illustrates a comparison between the symmetrically splayed spindles 602 and the angled spindles 701. Here, the spindle angle 705 is clearly shown as biasing toward the center of the end opening 104. Furthermore, the angled spindles 701 may be staggered or alternating in their length 704. Meaning, larger spindles 702 may be alternated with smaller spindles 703. By way of non-limiting example, the larger spindles 702 may all be the same length and the smaller spindles 703 may all be the same length. Furthermore, they may all share the same biasing angle toward the center.

FIG. 7C is a segmented environmental view of a first end 104 of the apparatus 100 with angled spindles 701. As shown, the tree branch 604 is clasped by the angled spindles 701 resulting in the apparatus 100 being secured in place. By way of non-limiting example, the angled spindles 701 may be bendable and not rigid in order to create a more custom and secure fit.

FIG. 8 begins to illustrate some of the functional features of the apparatus 100. As shown, the apparatus 100 is in an opened position 800. A tree branch 604 with a manually created fissure 803 is positioned along a medial longitudinal axis of the apparatus 100 between the first and second enclosing members 303a, 303b.

Exposed by the open position 800 is the apparatus body internal cavity 801 created by the cylindrical body base 103 and the body tapers 106 that house the tree branch 604. Also shown is an interior longitudinal convex ridge 802 formed by the longitudinal channel 201 on the exterior of the apparatus 100. Hence, each longitudinal channel 201 of the plurality of longitudinal channels 201 includes a correlating interior longitudinal convex ridge 802. As will be explained more fully hereafter, the plurality of interior longitudinal convex ridges 802 may be structured to create channel indentations within a root network mass formed within the internal cavity 801 of the apparatus 100.

FIG. 9 illustrates the apparatus 100 in a closed position 900 with a tree branch 604 positioned along the medial longitudinal axis of the apparatus 100 between the first and second enclosing members 303a, 303b within the internal cavity 801. The tree branch 604 is protruding through the first and second end openings 104, 105 while the apparatus 100 is secured shut by the plurality of joiner pegs 115 engaging the plurality of joiner holes 301. Therefore, the

first half enclosing member 303a and second half enclosing member 303b are shown as being structured to removably engage each other around a portion of a plant in a closed position 900 to form the whole apparatus 100.

FIGS. 10 and 11 illustrate how the apparatus 100 may further be secured around a tree branch 604 utilizing the fastening dowels 110. As shown, when in the closed position 900 a perimeter fastening strap 1001, which by way of non-limiting example may be a rubber band, may be secured around the fastening dowel 110 extending from the right side 111 of a first enclosing member 303a, wrapped around the apparatus 100 and secured to a fastening dowel 110 positioned on an inverted second enclosing member 303b. Therefore, the perimeter fastening strap 1001 is able to be wrapped around the exterior of the apparatus 100 further securing it in the closed position 900 around a tree branch 604.

FIG. 12 illustrates another securing means by which the apparatus 100 may be fastened in the closed position 900 around a tree branch 604. As shown, when in the closed position 900 the fastening apertures 114 located on the perimeter tab 107 may align. This alignment may facilitate the apparatus 100 being further secured closed around a tree branch 604 when a plurality of side cinches 1201 are threaded through the aligned fastening apertures 114. In some embodiments, the side cinches 1201 may be a standard cable tie.

FIG. 13 is a cross sectional environmental view of the apparatus 100 illustrating the newly formed roots 1301 that are created within the internal cavity 801. As shown, the tree branch 604 may be enclosed within the apparatus 100 and may extend through the first and second end openings 104, 105. After the manually created fissure 803 has been created, it may be enclosed with a potting mix 1302 such as peat moss, sphagnum moss, or the like. The tree branch 604 then begins to create the newly formed roots 1301, which ultimately results in a root network mass 1401.

FIGS. 14A and 14B are environmental views of the root network mass 1401 created by the apparatus 100. This root network mass 1401 is formed by a combination of a mature network of newly formed roots 1301 and either a sphagnum moss or potting mix 1302 that have been compacted by the apparatus 100. This root network mass 1401 is dense enough and compacted tightly enough within the apparatus 100 to have the plurality of interior longitudinal convex ridges 802 form a plurality of longitudinal indentions 1402 within the root network mass 1401.

FIG. 14B shows that the tree branch may be severed 1404 proximate one open end 105 so that it may be planted. It also illustrates that the longitudinal indentions 1402 may be on opposing sides of the root network mass 1401.

FIGS. 15A and 15B are environmental views of the root network mass 1401 and planted branch 1500 in its own separate soil 1501. They further show that the plurality of longitudinal indentions 1402 may be used as tracks to leverage stakes 1502 aligned therewith to supply support for the root network mass 1401 and base of the planted branch 1500. This may allow for the planted branch to grow upright and mature in a healthy uniform fashion. Similar to FIG. 14B, FIG. 15B illustrates that in some embodiments two stakes 1502 may align with the longitudinal indentions 1402 on opposing sides of the root network mass 1401.

FIGS. 16A through 16F illustrate different embodiments of the apparatus 100 that are contemplated to be within the scope of this application. As such, FIG. 16A shows an apparatus without body tapers 106. In this embodiment, open ends 104, 105 may be medially positioned within the

cylindrical body base **103**. FIG. **16B** illustrates that in some embodiments, an apparatus first end **101** may include a single body taper **1602** and an apparatus second end **102** may include only the truncated end **1603** of a cylindrical body base **103**. FIG. **16C** may be the same as the embodiment shown in FIG. **16A** with the addition of at least one longitudinal channel **201** positioned thereon. FIG. **16D** illustrates an embodiment whereby the cylindrical body base **103** includes inverted tapered ends **1605**. In this embodiment, the open ends **104**, **105** may be medially positioned between the inverted tapered ends **1602** within the cylindrical body base **103**. FIG. **16E** is similar to FIG. **16D**, however, it includes rounded vertices at the inverted tapered ends **1605**. Additionally, FIG. **16F** includes an embodiment whereby one end of the cylindrical body base **103** includes inverted tapered ends **1605** and the other includes a truncated end **1603**.

FIG. **17** illustrates another embodiment of the invention whereby the longitudinal channel **201** is supplemented with a plurality of lateral channels **1701**. The plurality of lateral channels **1701** may form a plurality of interior lateral convex ridges structured to create lateral channel indentions within a root network mass **1401** formed inside the apparatus **100**. These lateral channel indentions may assist with stabilizing and leveraging the root network mass **1401** when it is planted in its own separate soil **1501**. In other words, these lateral channel indentions within the root network mass **1401** may fill in with dirt when planted and may stabilize the planted branch **1500** as well as prevent it from easily being pulled out of its own separate soil **1501**.

FIGS. **18** and **19** illustrate another embodiment of the invention whereby the perimeter tab **107** may include elbow shaped T-levers **1804** structured to facilitate opening the apparatus **100** when both halves are engaged. Furthermore, this embodiment illustrates that the plurality of spindles **108** may include at least one spindle **108** shorter in length than the others **703** and at least one spindle end **109** distal to the end opening **1801** that is terminated by a cylinder **1805**. The spindles may also be angled **705** and in some embodiments, the plurality of spindles **108** on each half may include at least one pair of longer spindles **702** flanking at least one shorter medial spindle **703**.

Additionally, the perimeter tab **107** may include four rounded corners **1803** and a pair of parallel sides **1802**. Each side **1802** of each half may be completely surrounded or bordered by the perimeter tab **107**. At each longitudinal end **1801** of each half of the apparatus **100**, the perimeter tab **107** may converge into the end opening **1801**. In other words, the opposing end openings may form a concave void in the perimeter tab **107** at each respective end of the apparatus **100**. Furthermore, in some embodiments, the width of the body **150** compared to the end opening **1801** may equal 1 to 0.75 respectively. In some embodiments, the plurality of spindles **108** may include five (5) spindles **108** per half, per end. In other embodiments, the plurality of spindles **108** may include two (2) spindles **108** per half, per end. However, one skilled in the art will appreciate that the number of spindles **108** may be more or less depending on need, preference, and circumstance.

FIGS. **20A** through **20D** show the elbow shaped T-levers **1804** in more detail. As shown, they may include a pair of opposing T-flanks **2001** on either side of a drop member **2004**. The drop member **2004** may be perpendicular to an extender **2003** that protrudes from the perimeter tab **107** forming the elbow **2002**.

FIGS. **21** and **22** illustrates another embodiment of the invention whereby the perimeter tab **107** includes a finger

tab **2101** structured to facilitate a user opening the apparatus **100** when both halves are engaged. The finger tab **2101** may extend outward from the perimeter tab **107** and may include a u-shaped member designed to fit a standard user's finger within the u-shaped void.

FIGS. **23A** through **23D** illustrate the finger tab **2101** in more detail. As shown, the finger tab **2101** may include a pair of extending arms **2302** connected by a concave pull **2301**. The extending arms **2302** may cause the finger tab **2101** to extend distally from the perimeter tab **107** and the concave pull **2301** may be sized and dimensioned to allow the concave finger tab protrusion **2101** to facilitate a user's finger fitting into the created concave void. The concave pull **2301** may be used by a user to leverage and pull the perimeter tab **107** and apparatus half away from its engaged other half.

FIG. **24** illustrates another embodiment of the invention whereby opposing sides **1801** of the perimeter tab **107** may include a pair of fastening hooks **2401** structured to facilitate securing the apparatus **100** in a closed position with a perimeter fastening strap. The fastening hooks **2401** may abut an aperture **2403** within the perimeter tab **107**. This figure also emphasizes that in some embodiments there may be as few as two spindles **2402**. It also highlights that the spindle ends **109** may be terminated by cylinders **1805**.

FIGS. **25A** through **25C** emphasize the fastening hooks **2401** in more detail. As shown, they may be positioned on the perimeter tab **107** and may include a hook housing **2501**, a hook hood **2502**, and a hook void **2503** within the hook housing. The hook hood **2502** may allow for a fastening strap to remain in place without prolapsing or slipping off entirely.

That which is claimed is:

1. A tree rooting enclosure apparatus comprising:

a body comprising:

a cylindrical body base,

a pair of opposing end openings with edges exterior to the body; and

a plurality of spindles longitudinally oriented with respect to the cylindrical body base and extending outward from the exterior edges of the pair of opposing end openings;

wherein the tree rooting enclosure apparatus is comprised of two removably engageable halves, each half being an identical duplicate of the other;

wherein each half is configured to mate with the other half when each respective half is inversely oriented with respect to the other half; and

wherein both halves when engaged envelop at least a portion of a tree branch to propagate a rooting system.

2. The tree rooting enclosure apparatus of claim 1 wherein the plurality of spindles comprises at least one spindle shorter in length than the remaining spindles.

3. The tree rooting enclosure apparatus of claim 1 wherein the spindles are configured in at least one of a symmetrically splayed orientation and an angled orientation.

4. The tree rooting enclosure apparatus of claim 1 wherein the cylindrical body base diameter is proportional to the end opening diameters by a ratio of 1 to 0.75 respectively.

5. The tree rooting enclosure apparatus of claim 1 wherein the cylindrical body base comprises a perimeter tab extending distally therefrom.

6. The tree rooting enclosure apparatus of claim 5 wherein the perimeter tab comprises an elbow shaped T-lever configured to facilitate opening the tree rooting enclosure apparatus when both halves are engaged.

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7. The tree rooting enclosure apparatus of claim 5 wherein the perimeter tab comprises four rounded corners and a pair of parallel sides.

8. The tree rooting enclosure apparatus of claim 5 wherein the pair of opposing end openings form a concave void in the perimeter tab at each respective end of the tree rooting enclosure apparatus.

9. A tree rooting enclosure apparatus comprising:

a body comprising:

a cylindrical body base,

a perimeter tab extending from the cylindrical body base, and

a pair of opposing end openings with edges exterior to the body;

a plurality of spindles longitudinally oriented with respect to the cylindrical body base and extending outward from the exterior edges of the pair of opposing end openings; and

a plurality of longitudinal channels extending the length of the cylindrical body base;

wherein the tree rooting enclosure apparatus is comprised of two removably engageable halves, each half being an identical duplicate of the other;

wherein each half is configured to mate with the other half when each respective half is inversely oriented with respect to the other half; and

wherein both halves when engaged envelop at least a portion of a tree branch to propagate a rooting system.

10. The tree rooting enclosure apparatus of claim 9 wherein the plurality of spindles comprises at least one pair of longer spindles flanking at least one shorter medial spindle.

11. The tree rooting enclosure apparatus of claim 9 wherein opposing sides of the perimeter tab comprise a pair of fastening hooks configured to facilitate securing the tree rooting enclosure apparatus in a closed position with a perimeter fastening strap.

12. The tree rooting enclosure apparatus of claim 9 wherein at least one of the plurality of spindles comprises an end distal to its respective end opening that is terminated by a cylinder.

13. The tree rooting enclosure apparatus of claim 9 wherein the perimeter tab of each half includes a concave protrusion configured to facilitate a user pulling apart the two halves when the two halves are engaged.

14. The tree rooting enclosure apparatus of claim 13 wherein the concave protrusion is dimensioned to allow a user's finger to fit into its concave void.

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15. The tree rooting enclosure apparatus of claim 9 wherein the perimeter tab comprises a plurality of fastening apertures configured to accommodate a plurality of side cinches for securing the tree rooting enclosure apparatus in a closed position.

16. The tree rooting enclosure apparatus of claim 9 wherein the perimeter tab comprises at least one side with a plurality of joinder pegs and an opposing side with a plurality of joinder holes.

17. The tree rooting enclosure apparatus of claim 16 wherein the plurality of joinder pegs of the first half enclosing member is configured to removably engage the plurality of joinder holes of the second half enclosing member to form a water-resistant fitting when the apparatus is in a closed position.

18. The tree rooting enclosure apparatus of claim 9 wherein the spindles are in an angled orientation.

19. The tree rooting enclosure apparatus of claim 9 wherein the plurality of spindles includes at least one spindle shorter in length than the others and at least one spindle comprises an end distal to its respective end opening that is terminated by a cylinder.

20. A tree rooting enclosure apparatus comprising:

a body comprising:

a cylindrical body base,

a perimeter tab extending the entire longitudinal length of two sides of the cylindrical body base comprising at least one of a concave finger tab extension and an elbow shaped T-lever,

a pair of opposing end openings with edges exterior to the body;

a plurality of spindles longitudinally oriented with respect to the cylindrical body base and extending outward from the exterior edges of the pair of opposing end openings;

wherein the tree rooting enclosure apparatus is comprised of two removably engageable halves, each half being an identical duplicate of the other;

wherein each half is configured to mate with the other half when each respective half is inversely oriented with respect to the other half;

wherein both halves when engaged envelop at least a portion of a tree branch to propagate a rooting system; and

wherein the concave finger tab protrusion and elbow shaped T-lever are configured to facilitate a user pulling apart the two halves when the two halves are engaged.

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